



TIA STANDARD

Project 25

Link Control Word Formats and Messages Addendum 2- User Alias

TIA-102.AABF-D-2
(Addendum to TIA-102.AABF)

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(This foreword is not part of this addendum.)

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard.

This document has been developed with inputs from the APCO/TIA Project 25 Interface Committee (APIC), the Trunking and Conventional Control Subcommittee (TR-8.10), and TIA Industry members.

This document describes enhancements to the link control formats and messages for conventional and trunked systems utilizing the Common Air Interface

As other documents describing the conventional and trunking protocols are developed, this document will be subject to change. All reasonable efforts have been made to ensure accuracy, but it should be understood that conformance test remain to be developed and proven.

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ADDENDUM INTRODUCTION

The purpose of this addendum is to update information contained in TIA-102.AABF-D, "Link Control Word Formats and Messages". Details are provided in the "Addendum Scope" section below.

Addendum Scope

This addendum provides the following enhancements to the Link Control Word Formats and Messages (TIA-102. AABF-D) standard:

1. Adds new messages for the User Alias functionality.

This document only includes information relevant to the scope as described above. The reader is referred to TIA-102.AABF-D for all other pertinent information. The next complete revision of TIA-102.AABF-D will incorporate the changes contained in this addendum.

Note that section numbers identified in this addendum directly translate to section numbers in the TIA-102.AABF-D, Link Control Word Formats and Messages.

Addendum Revision History

Version	Date	Description
TIA-102.AABF-D-2	June, 2023	Approved for publication

Addendum changes to TIA-102.AABF-D, Link Control Word Formats and Messages, begin here.

[Changes begin as identified below. No changes from the existing text before this clause]

7.3 LC Messages

7.3.35 User Alias Part A

This LC is used outbound during an FDMA call to convey the first part, Part A, of a subscriber unit user's user alias. The user alias of the subscriber is determined by combining this LC with the User Alias Part B. Combination of Parts A and B from the same conveyance yield a 14-character field to convey user alias on the outbound channel. Each ASCII character shall be transmitted over the air.

Octet 0	P	SF	LCO						LCO = 50 (%11,0010), or 52 (%11,0100) P = 0, SF = 0
1	Manufacturer ID								
2	ASCII Character(0)								
3	ASCII Character (1)								
4	ASCII Character (2)								
5	ASCII Character (3)								
6	ASCII Character (4)								
7	ASCII Character (5)								
8	ASCII Character (6)								
Bit	7	6	5	4	3	2	1	0	

LC Opcode: 50(%11,0010) for odd-numbered conveyance (1st, 3rd, 5th).
52(%11,0100) for even-numbered conveyance (2nd, 4th, 6th).

For odd-numbered conveyances, including the first message, the radio uses LCO=50 for Part A and LCO=51 for Part B.

For even-numbered conveyances, the radio uses LCO=52 for Part A and LCO=53 for Part B.

Manufacturer ID: Set to \$A4

ASCII Character (0 through 6): The first 7 bytes of the ASCII character user alias. If Alias is '123456789abcde' then Part A = '1234567'. Unused characters are set to null.

7.3.36 User Alias Part B

This LC is used outbound during an FDMA call to convey the second part, Part B, of a subscriber unit user's user alias. The user alias of the subscriber is determined by combining this LC with the User Alias Part A. Combination of Parts A and B from the same conveyance yield a 14-character field to convey user alias on the outbound channel. Each ASCII character shall be transmitted over the air.

Octet 0	P	SF	LC Opcode					LCO = 51 (%11,0011), or 53 (%11,0101) P = 0, SF = 0
1	Manufacturer ID							
2	ASCII Character (7)							
3	ASCII Character (8)							
4	ASCII Character (9)							
5	ASCII Character (10)							
6	ASCII Character (11)							
7	ASCII Character (12)							
8	ASCII Character (13)							
Bit	7	6	5	4	3	2	1	0

LC Opcode: 51(%11,0011) for odd-numbered conveyance (1st, 3rd, 5th).
53(%11,0101) for even-numbered conveyance (2nd, 4th, 6th).

For odd-numbered conveyance, including the first conveyance, the radio uses LCO=50 for Part A and LCO=51 for Part B.

For even-numbered conveyance, the radio uses LCO=52 for Part A and LCO=53 for Part B.

Manufacturer ID: Set to \$A4

ASCII Character (7 through 13): The second 7 bytes of the ASCII character User Alias. If User Alias is '123456789abcde' then Part B = '89abcde'. Unused characters are set to null.

7.4 Standard LC Field Definitions

[Added Link Control Opcodes]

Add to Table 2: LCO – Link Control Opcode

LCO Value	Alias Name	Full Name Mnemonic
%110010	U_ALIAS_A_ODD	User Alias Part A – Odd-numbered conveyance
%110011	U_ALIAS_B_ODD	User Alias Part B – Odd-numbered conveyance
%110100	U_ALIAS_A_EVEN	User Alias Part A – Even-numbered conveyance
%110101	U_ALIAS_B_EVEN	User Alias Part B – Even-numbered conveyance

7.5 Link Control Word Usages

[Added Link Control Opcodes]

Add to Table 3: Link Control Word Usage



LC Function	LCO Value				
User Alias Part A – Odd-numbered conveyance	%110010		x		
User Alias Part B – Odd-numbered conveyance	%110011		x		
User Alias Part A – Even-numbered conveyance	%110100		x		
User Alias Part B – Even-numbered conveyance	%110101		x		

[End of changes]

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1310 N. COURTHOUSE ROAD, SUITE 890
ARLINGTON VA, 22201; USA

+1.703.907.7700 MAIN
+1.703.907.7727 FAX